

Air Quality Analysis and Prediction Using Machine Learning Models

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ABSTRACT

This study proposes a machine learning-driven framework to predict air quality levels across five Ontario cities: Windsor, Hamilton Downtown, London, Sarnia, and Toronto West. Historical hourly data on air pollutants and meteorological variables from 2021 to 2023 were utilized to build ensemble models including XGBoost, LightGBM, CatBoost, and Gradient Boosting for estimating the Air Quality Health Index (AQHI). The analysis was conducted in two phases: regression models for continuous AQHI prediction and classification models for categorical AQHI levels (Low, Moderate, High). Incorporating weather features, such as temperature, led to slight but consistent improvements in model performance across all cities, with XGBoost classification accuracy increasing by up to 6% in Windsor and 1–3% elsewhere. Classification models, especially CatBoost and LightGBM, achieved the highest accuracy, reaching up to 90% in certain locations. Additionally, ARIMA models were used to forecast monthly pollutant concentrations (PM_{2.5}, NO₂, O₃), using 2021–2022 as the training period and 2023 as the forecast window. While seasonal patterns for O₃ and NO₂ were reasonably captured, PM_{2.5} forecasts remained less reliable, suggesting the need for more advanced time-series models. These results demonstrate the value of incorporating meteorological features and pollutant forecasting for more accurate and actionable AQHI predictions in public health planning.

Keywords: Air Quality, AQHI, XGBoost, CatBoost, LightGBM, Gradient Boosting, Classification, Weather Integration

INTRODUCTION

Air pollution continues to be a pressing issue for urban regions, influencing public health and environmental policy. This study aims to address this concern by developing a scalable machine learning framework for predicting air quality across multiple Ontario cities using historical pollution and weather data.

1.1 Objective

The objective of this project is to develop machine learning models to predict air quality levels across five Ontario cities: Windsor, Hamilton Downtown, London, Sarnia, and Toronto West. The study aims to assess the effectiveness of ensemble models for both regression and classification tasks, evaluate the influence of meteorological variables on prediction accuracy, and provide a framework for reliable AQHI forecasting.

1.2 Scope

This work builds on a previous analysis conducted for Windsor and extends the methodology to four additional cities with distinct pollution profiles. The scope includes data collection from public sources, preprocessing of hourly pollutant and weather data from 2021 to 2023, exploratory data analysis, and the implementation of regression and classification models. The study further explores the effect of including weather features on model performance and compares multiple machine learning algorithms for their suitability in air quality prediction.

RELATED WORK

The prediction of air quality using machine learning techniques has gained increasing relevance due to the pressing need for accurate environmental forecasts. Numerous studies have underscored the influence of both local and regional pollution sources, and the value of integrating environmental and meteorological data in predictive modeling.

Zhang et al. [1] conducted a temporal analysis of AQHI trends in Windsor, Ontario, from 2016 to 2019. The study identified significant seasonal variation in pollutant contributions, with ozone (O₃) playing a major role in the

warm season and nitrogen dioxide (NO₂) dominating during colder months. The analysis also pointed to the impact of transboundary pollution from the industrial regions of the United States, highlighting the complexity of urban air quality dynamics.

Wang et al. [2] evaluated the integration of meteorological variables such as temperature and humidity into air quality prediction models in Beijing. Their findings revealed that such integration significantly improved predictive performance, especially during pollution episodes driven by atmospheric stagnation. This study demonstrated the importance of accounting for meteorological influences when modeling pollutant behavior.

Liu et al. [3] applied a combination of regression and neural network models to forecast PM_{2.5} and ozone concentrations in Los Angeles. Incorporating traffic and weather data enabled the models to better capture temporal spikes and dispersion patterns. This hybrid methodology supported the argument for multimodal data integration in air quality forecasting systems.

Ravindiran et al. [4] tested several ensemble models, including Random Forest, XGBoost, LightGBM, CatBoost, and AdaBoost, on a multi-year dataset in Visakhapatnam, India. The study revealed that LightGBM and Random Forest offered strong predictive power and interpretability. It also emphasized the advantage of combining pollutant and meteorological features for improved model generalization.

These existing solutions and research findings validate the use of ensemble-based machine learning models and confirm the relevance of meteorological factors in air quality prediction. They serve as a theoretical and empirical foundation for the present study, which seeks to implement and compare similar approaches across five Ontario cities.

METHODOLOGY

3.1 Problem Formulation

This study aims to predict AQHI values across five Ontario cities using machine learning models. The prediction task is framed in two parts: regression for estimating continuous AQHI values and classification for assigning AQHI to categorical levels (Low, Moderate, High). The performance of both modeling approaches is evaluated, along with the impact of adding meteorological features.

3.2 Data Collection and Preprocessing

Hourly data on air pollutants (PM_{2.5}, SO₂, NO₂, CO, NO, and O₃) were obtained from Air Quality Ontario [5], while meteorological data (temperature and humidity) were sourced from Environment Canada [6]. The dataset spans January 2021 to December 2023.

Key preprocessing steps:

- Replaced outliers (e.g., -999, 9999) with local means
- Filled missing values using rolling averages
- Created time-based features (hour, day, month)
- Merged pollutant and weather datasets using timestamp alignment

3.3 Exploratory Data Analysis (EDA)

EDA showed distinct temporal and seasonal pollutant patterns:

- PM_{2.5} and NO₂ peaked during rush hours; O₃ peaked midday.
- Summer AQHI values were higher; winter values mostly fell in the Low category.
- Windsor and Hamilton had more pronounced pollution levels; Toronto West remained stable.
- O₃ and NO₂ exhibited an inverse relationship consistent with known atmospheric processes.

These patterns supported the decision to engineer time features and test weather integration.

3.4 Model Development

The modeling process was conducted in a structured, multi-stage approach to predict the Air Quality Health Index (AQHI) across five Ontario cities. The goal was to determine the most effective algorithms and input feature sets for both regression and classification tasks.

Feature Selection Criteria

Feature selection was guided by a correlation matrix analysis. Only variables with a correlation coefficient greater than 0.3 with AQHI were retained to ensure relevance. As a result, two feature sets were formed:

- Pollutants Only: 'PM2.5', 'O3', 'NO2'
- Pollutants + Weather: 'PM2.5', 'O3', 'NO2', 'Temp (°C)'

Three-Stage Framework

Stage 1: Regression (Pollutants Only)

In the first stage, the ensemble models selected through preliminary testing—XGBoost, LightGBM, CatBoost, and Gradient Boosting—were trained using only pollutant data: PM2.5, O₃, and NO₂. The goal in this phase was to predict AQHI as a continuous numerical value, which served as a baseline for evaluating the impact of additional meteorological variables.

Stage 2: Regression (Pollutants + Weather)

The second stage extended the input feature set to include temperature (°C), the only meteorological variable meeting the correlation threshold. The objective was to assess whether integrating weather-related data could enhance the model's ability to predict AQHI continuously. This stage followed the same regression framework as Stage 1 but with a broader feature set: ['PM2.5', 'O3', 'NO2', 'Temp (°C)'].

Stage 3: Classification (AQHI Categories)

To improve interpretability and provide more actionable public health insights, the third stage reformulated AQHI prediction as a classification problem. AQHI values were categorized into three health-relevant levels: **Low** (1–3), **Moderate** (4–6), **High** (7 and above)

The same ensemble models were applied to this classification task using both the pollutant-only and pollutant-plus-temperature feature sets. This phase allowed the comparison of model effectiveness in categorical prediction scenarios, where classification accuracy becomes the key metric

RESULTS

The evaluation of model performance was carried out using key metrics such as Accuracy, Root Mean Squared Error (RMSE), R² score, for regression, and classification accuracy. Results were analyzed to assess the impact of weather data and the comparative strength of regression versus classification approaches.

4.1 Regression Using Pollutant-Only Features

In the first stage of modeling, eight regression algorithms were evaluated using only pollutant variables (PM2.5, O₃, NO₂). The goal was to predict AQHI as a continuous variable and identify the most suitable models for further analysis.

Figure 1 presents the test accuracy of each model across the five cities. Ensemble methods such as XGBoost, CatBoost, LightGBM, Gradient Boosting, and Random Forest consistently outperformed simpler models like Linear Regression, Decision Tree, and AdaBoost.

- XGBoost achieved the highest accuracy in most cities, especially in London and Sarnia, confirming its robustness.
- CatBoost and Gradient Boosting also showed competitive and stable performance across all locations.
- Simpler models like Decision Tree and AdaBoost underperformed, particularly in Windsor and Hamilton Downtown.

Based on this comparison, XGBoost, LightGBM, CatBoost, and Gradient Boosting were selected for deeper analysis in subsequent stages (regression with weather and classification).

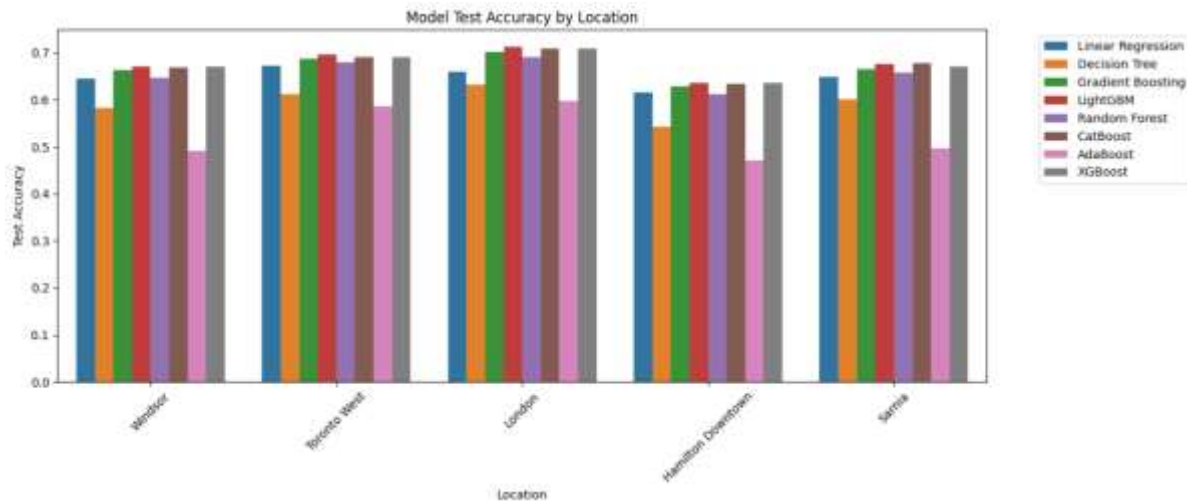


Figure 1 Regression model test accuracy by location

4.2 Regression with Weather Features

In Stage 2, temperature (Temp (°C)) was added to the pollutant-only feature set (PM2.5, O₃, NO₂) to evaluate the effect of meteorological data on regression performance. XGBoost, selected from Stage 1, was used as the base model due to its consistent performance.

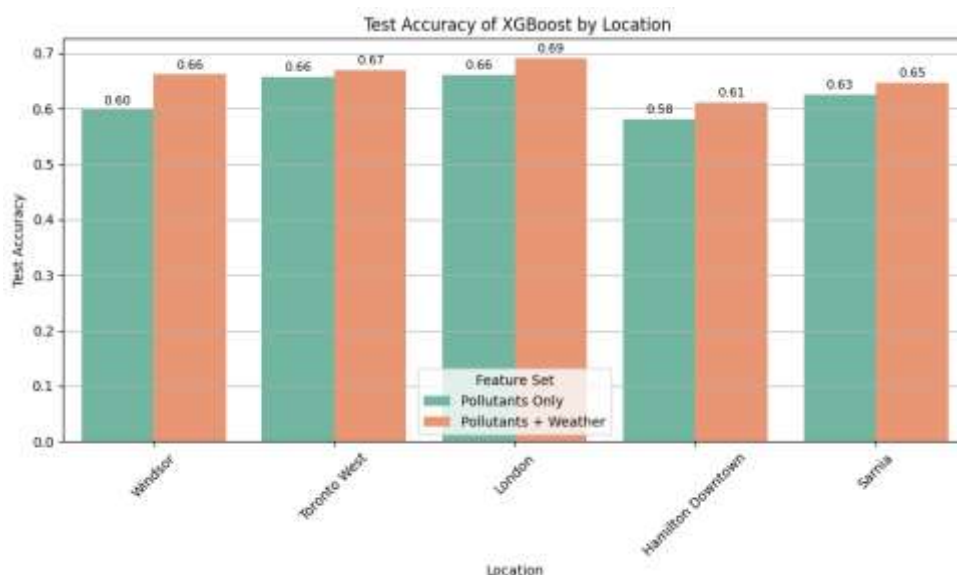


Figure 2 Test accuracy of XGBoost by location

Test Accuracy improved modestly across all five cities with the inclusion of temperature (Figure 2).

- Windsor showed the largest gain ($0.60 \rightarrow 0.66$), followed by London ($0.66 \rightarrow 0.69$) and Hamilton Downtown ($0.58 \rightarrow 0.61$).
- Gains in Toronto West and Sarnia were smaller but positive.

RMSE (Figure 3) decreased with weather integration, indicating better predictive accuracy.

- London ($0.59 \rightarrow 0.55$) and Windsor ($0.67 \rightarrow 0.60$) showed the most notable improvements.
- All cities saw lower error with temperature included.

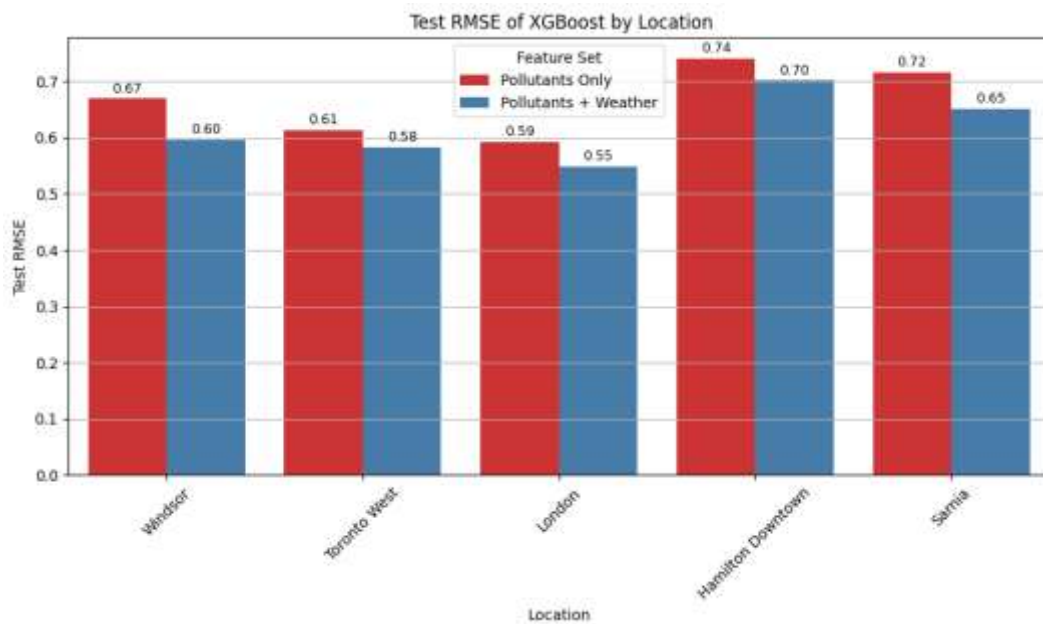


Figure 3 Regression RMSE comparison

R² Scores also increased in every location (Figure 4), reflecting improved model fit.

- Sarnia ($0.45 \rightarrow 0.55$), London ($0.39 \rightarrow 0.48$), and Hamilton Downtown ($0.41 \rightarrow 0.47$) benefited significantly from the additional feature.

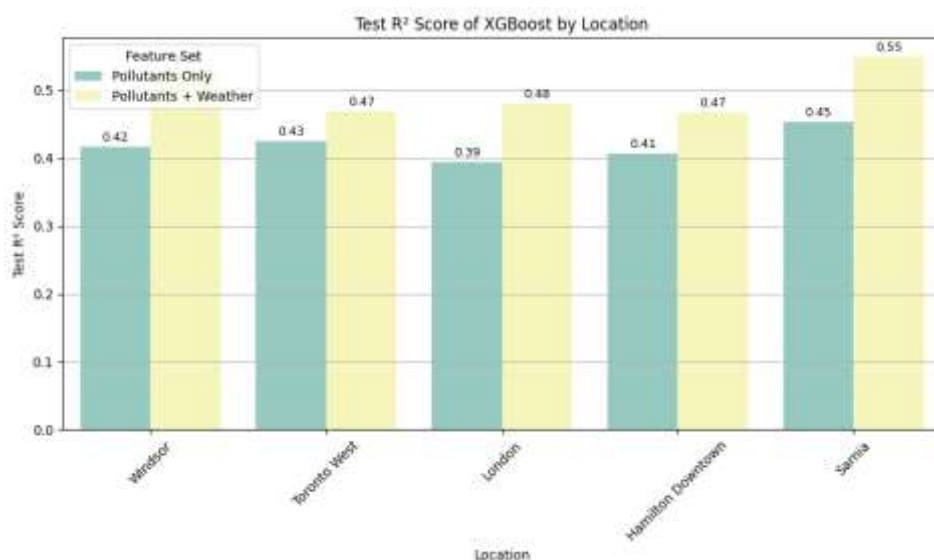


Figure 4 Regression R² comparison

These results confirm that adding temperature contributed to overall performance enhancement—though improvements were modest, they were consistent across metrics and cities.

4.3 Classification Results

In Stage 3, Classification models outperformed regression models in consistency and accuracy. As shown in Figure 5, all models achieved high and stable accuracy across both feature sets:

- XGBoost, LightGBM, and CatBoost each maintained a consistent accuracy of 0.90, regardless of whether temperature was included.
- Random Forest showed a slight improvement, increasing from 0.87 (pollutants only) to 0.90 with weather data.

These results confirm that classification models outperformed regression models in both consistency and reliability. Moreover, the addition of temperature did not significantly affect classification performance, indicating that AQHI categories can be effectively predicted using pollutant data alone.

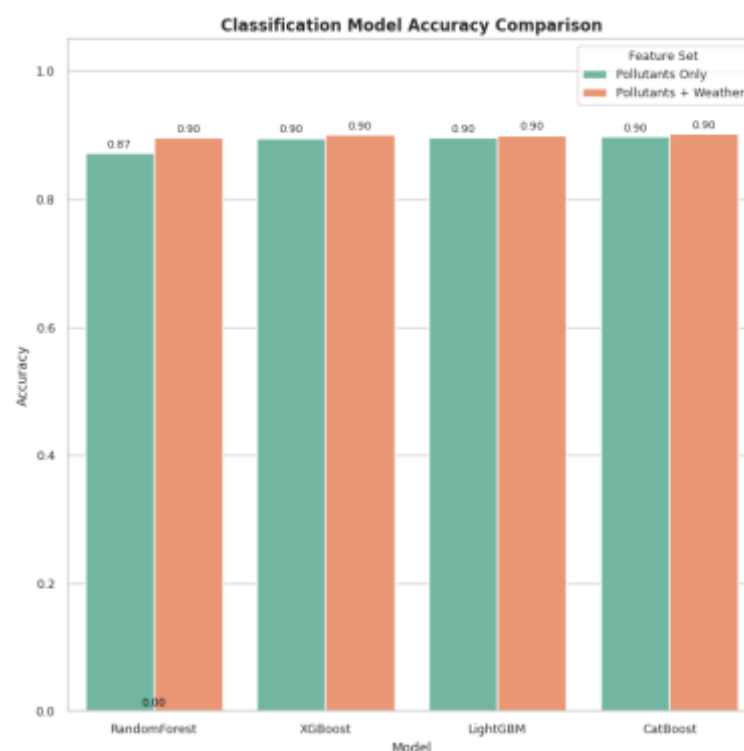


Figure 5 Classification accuracy comparison with weather data

4.3 ARIMA Forecasting for Pollutants

An ARIMA-based time-series forecast was performed for PM_{2.5}, O₃, and NO₂, using 2021–2022 data for model training and the full year of 2023 for validation (see Figure 6).

- **O₃ and NO₂:** Forecasts closely followed the observed seasonal trends in 2023, demonstrating that ARIMA effectively captures periodic pollutant cycles. This promising result suggests these data may serve as valuable inputs for future AQHI prediction systems.
- **PM_{2.5}:** ARIMA struggled to mirror the sharp peaks seen in mid-2023. However, this is likely due to unprecedented wildfire smoke events in Ontario, rather than model inadequacy. In fact, 2023 experienced record-breaking wildfires across Canada, over 18 million hectares burned, with Ontario impacted by two major smoke episodes in June [7][8]. Such extreme anomalies are beyond what traditional ARIMA can model.

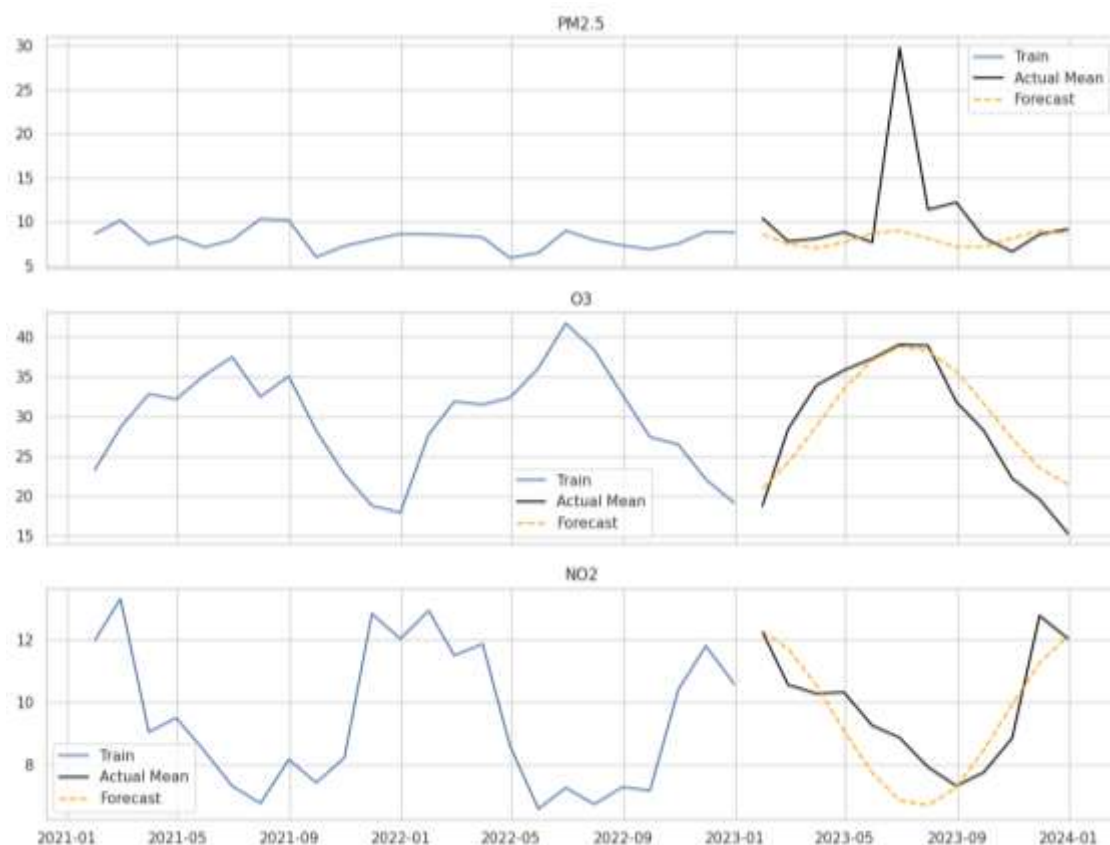


Figure 6 Monthly ARIMA Forecasting

DISCUSSION

The results of this study demonstrate that classification models are better suited for AQHI prediction than regression models, owing to their consistently higher accuracy and greater stability across all five cities. Among the classifiers tested, CatBoost, LightGBM, and XGBoost achieved accuracies up to 90%, making them ideal for public health applications where clear, actionable outputs are preferred.

While weather features, specifically temperature were initially expected to enhance model performance, their contribution was modest and location dependent. In regression tasks, temperature led to slight improvements in RMSE and R^2 scores in all cities, particularly London, Sarnia, and Windsor. However, in classification models, the addition of weather data made little to no difference, indicating that pollutant concentrations alone are sufficient for categorical AQHI prediction in the short term.

The variation in performance across cities highlights the importance of local environmental factors, such as industrial activity and traffic patterns. For instance, Hamilton Downtown and Sarnia, both home to major industrial operations, showed more volatile AQHI patterns. These findings support the need for location-specific modeling strategies to improve air quality prediction.

In addition, ARIMA-based forecasting showed encouraging results, particularly for O_3 and NO_2 , where the seasonal patterns were well captured. However, $PM_{2.5}$ forecasts underperformed, a shortcoming likely attributable to the extreme wildfire smoke events in Ontario during mid-2023, which produced unpredictable spikes in pollutant levels. These anomalies, not modeled in historical trends, suggest the need for advanced time-series approaches such as SARIMA, LSTM, or hybrid models capable of handling such irregular events [7][8].

Finally, the ability of classification models to categorize AQHI into Low, Moderate, and High-risk levels reinforces their utility for real-world deployment in health alert systems, public dashboards, and mobile apps.

5.1 Future Scope

While the current models worked well for predicting AQHI levels, forecasting individual pollutants still needs improvement. In the future, time-series methods like ARIMA can be used to better predict pollutant levels using past data. These predictions can be given as inputs to the AQHI models, possibly improving their accuracy.

Other ideas for future work include adding real-time prediction, using satellite data, or testing deep learning models. These could help cover more areas and make the system more reliable for use in public health and planning.

CONCLUSION

This project presented a comprehensive machine learning framework for predicting AQHI levels across five Ontario cities using pollutant and meteorological data. A wide range of models were initially evaluated, with XGBoost, LightGBM, CatBoost, and Gradient Boosting selected for in-depth analysis. Regression models benefited slightly from the inclusion of temperature data, particularly in cities like London and Sarnia, as evidenced by reduced RMSE and improved R^2 . However, classification models outperformed regression consistently, achieving 90% accuracy across all test cases, and weather features had negligible impact in this setting. Pollutant-driven models proved sufficient for reliable AQHI categorization, and their performance remained robust even when simplified to exclude meteorological variables.

It was also observed that local factors such as industrial activity and traffic congestion significantly influenced AQHI variability across different cities. For instance, Sarnia and Hamilton Downtown, known for their industrial footprint, exhibited more volatile AQHI patterns. Understanding these local contributions can aid in tailoring prediction models and public health responses more effectively. Overall, the models were able to capture meaningful patterns from pollutant data and showed strong classification accuracy. These findings can support the use of ensemble classifiers in real-world air quality monitoring systems where clear health alerts are more beneficial than precise numerical outputs.

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